the philosopher Thomas Nagel points out that a Sapiens mind cannot fathom the subjective world of a bat. We can write all the algorithms we want about the bat body, about bat echolocation systems and about bat neurons, but it won't tell us how it *feels* to be a bat. How does it feel to echolocate a moth flapping its wings? Is it similar to seeing it, or is it something completely different?

Trying to explain to a Sapiens how it feels to echolocate a butterfly is probably as pointless as explaining to a blind mole how it feels to see a Caravaggio. It's likely that bat emotions are also deeply influenced by the centrality of their echolocation sense. For Sapiens, love is red, envy is green and depression is blue. Who knows what echolocations colour the love of a female bat to her offspring, or the feelings of a male bat towards his rivals?

Bats aren't special, of course. They are but one out of countless possible examples. Just as Sapiens cannot understand what it's like to be a bat, we have similar difficulties understanding how it feels to be a whale, a tiger or a pelican. It certainly feels like something; but we don't know like what. Both whales and humans process emotions in a part of the brain called the limbic system, yet the whale limbic system contains an entire additional part which is missing from the human structure. Maybe that part enables whales to experience extremely deep and complex emotions which are alien to us? Whales might also have astounding musical experiences which even Bach and Mozart couldn't grasp. Whales can hear one another from hundreds of kilometres away, and each whale has a repertoire of characteristic 'songs' that may last for hours and follow very intricate patterns. Every now and then a whale composes a new hit, which other whales throughout the ocean adopt. Scientists routinely record these hits and analyse them with the help of computers, but can any human fathom these musical experiences and tell the difference between a whale Beethoven and a whale Justin Bieber?[4]